

A Synopsis

On

HYBRID INTRUSION DETECTION SYSTEM

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By

Akshat shakya (CS-2341119)

Kushal Sharma (CS-23411110)

Ruprekha behera (CS-23411595)

Shivam (CS-2341121)

Vivek Chaudhary (CS-2341433)

Under the guidance of

Dr. Narinder Verma



IILM UNIVERSITY,

16-18, Knowledge Park II,

Greater Noida

Uttar Pradesh 201306



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING IILM UNIVERSITY, GREATER NOIDA

Title :

Hybrid Intrusion Detection System Using Anomaly and Signature-Based Techniques

Introduction:

Signature-based IDS fails to detect zero-day attacks while anomaly-based IDS generates high false positives. A hybrid system is needed to detect both known and unknown attacks with reduced false alarms.

Problem Statement:

Signature-based IDS fails to detect zero-day attacks while anomaly-based IDS generates high false positives. A hybrid system is needed to detect both known and unknown attacks with reduced false alarms.

Objectives:

To design and implement a hybrid intrusion detection system that combines anomaly-based and signature-based techniques for improved attack detection and reduced false alarm rates.

Methodology:

Traffic is preprocessed and passed through a signature-based module for known threats. Remaining traffic goes to an ML-based anomaly detection module trained on NSL-KDD and UNSW-NB15. A decision engine combines both outputs to generate final alerts.

Tools and Technologies:

Python, Scikit-learn, Random Forest, SVM, NSL-KDD, UNSW-NB15, Snort, Wireshark, Jupyter Notebook.

Expected Outcomes:

Higher detection accuracy, reduced false positives, real-time detection of known and unknown threats through a unified hybrid framework.

Application:

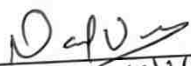
Enterprise Network Security, Cloud Infrastructure, Data Centres, Educational Networks, Industrial Control Systems

Conclusion:

Enterprise Network Security, Cloud Infrastructure, Data Centres, Educational Networks, Industrial Control Systems

Date: 23/3/24

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TITLE OF THE RESEARCH PROPOSAL	Hybrid Intrusion Detection System Using Anomaly and Signature-Based Techniques
NAME OF THE STUDENTS	Akshat Shakya ,Kushal Sharma Ruprekha Behera , Shivam, Vivek Chaudhary
NAME AND DESIGNATION OF THE GUIDE	Mr. Narinder verma, Assistant Professor
PLACE OF WORK	IILM University, Greater Noida
AREA	Cyber Security
DOMAIN	Network Security
CO-DOMAIN	Machine Learning / Data Analytics
APPLICATION AREA	Enterprise Network Security, Cloud Infrastructure, Data Centre Protection
USEFULNESS	Improves detection accuracy by combining known attack detection with unknown threat identification.
ANY KEY INFORMATION	The system aims to reduce false positives while enhancing real-time intrusion detection.


 (Dr. Narinder Verma)
 23/03/2021

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Statement of the proposal

Problem Statement

Traditional intrusion detection systems rely either on signature-based detection or anomaly-based detection. Signature-based systems fail to detect zero-day attacks, while anomaly-based systems often generate high false positives. There is a need for a more robust system that can effectively detect both known and unknown attacks with improved accuracy.

Background

With the rapid growth of internet-based services, networks have become vulnerable to various cyber-attacks. Intrusion Detection Systems (IDS) play a critical role in monitoring network traffic and identifying malicious activities. However, existing IDS techniques have limitations when used independently.

Motivation/need for the research:

- Increasing frequency of sophisticated cyber-attacks
- Limitations of standalone IDS approaches
- Requirement for enhanced accuracy and reduced false alarms
- Growing importance of proactive cyber defense mechanisms
- High false alarm rates in existing systems causing alert fatigue for security teams

Objective

To design and implement a hybrid intrusion detection system that combines anomaly-based and signature-based techniques for improved attack detection.

Sub-Objectives

- To analyze existing IDS techniques
- To design an anomaly detection module using machine learning
- To integrate signature-based detection for known attacks
- To evaluate system performance using standard datasets
- To minimize false positive alerts by combining outputs through a decision engine

Mode of achieving objective

- Study of existing IDS architectures.
- Selection of suitable machine learning algorithms.
- Dataset preprocessing and feature extraction.
- System implementation and testing.

Proposed Methodology

The proposed system will first analyze network traffic using a signature-based detection module to identify known threats. Traffic not matching known signatures will be passed to an anomaly detection module trained using machine learning techniques. Alerts generated by both modules will be combined to produce final intrusion detection results.

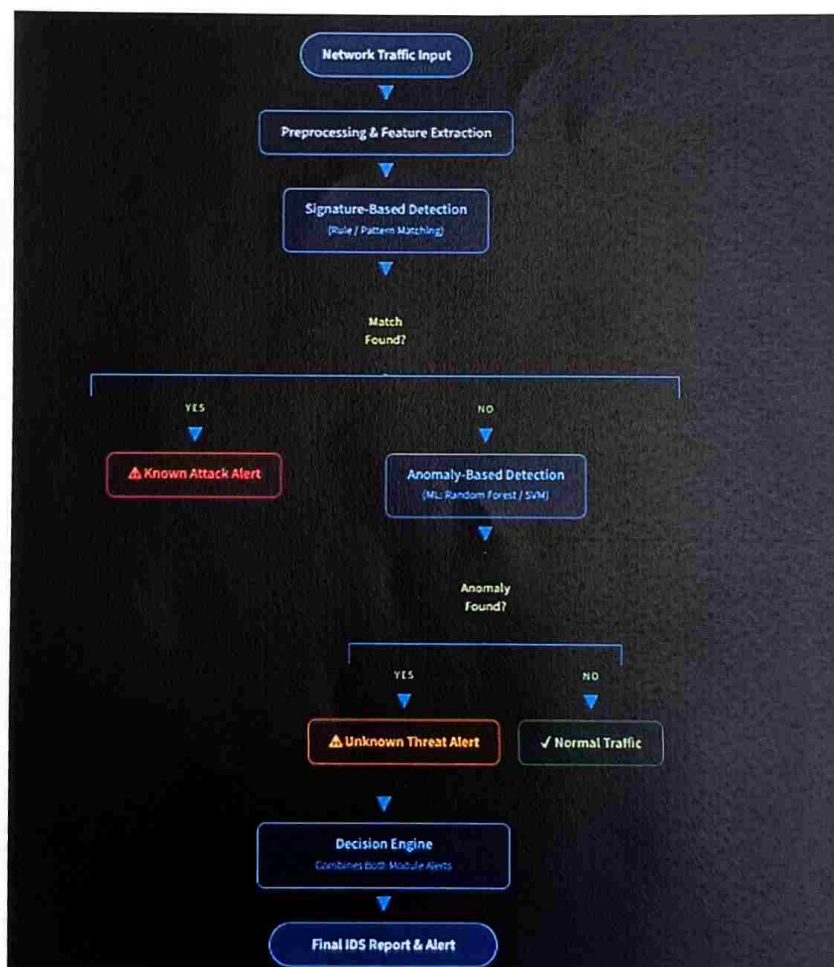
Theoretical framework

The framework is based on combining rule-based detection (signature IDS) with behaviour-based detection (anomaly IDS). Machine learning models learn normal traffic patterns, while predefined signatures identify known attack vectors.

Sources of data

Secondary data will be used, including benchmark datasets such as NSL-KDD or UNSW-NB15 for training and testing.

Figure 1. *Work flow Diagram (Clarity on developing System/Method/Model/Platform/Apparatus)*



Schematic flow Diagram:

Figure 2. *Schematic Flow Diagram*

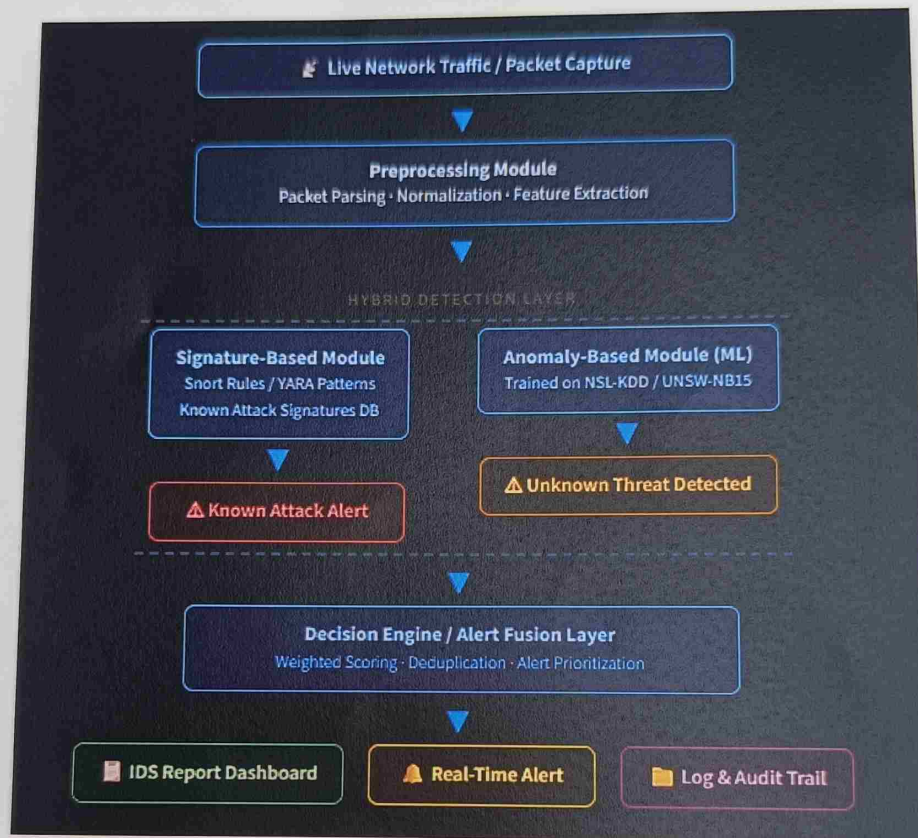
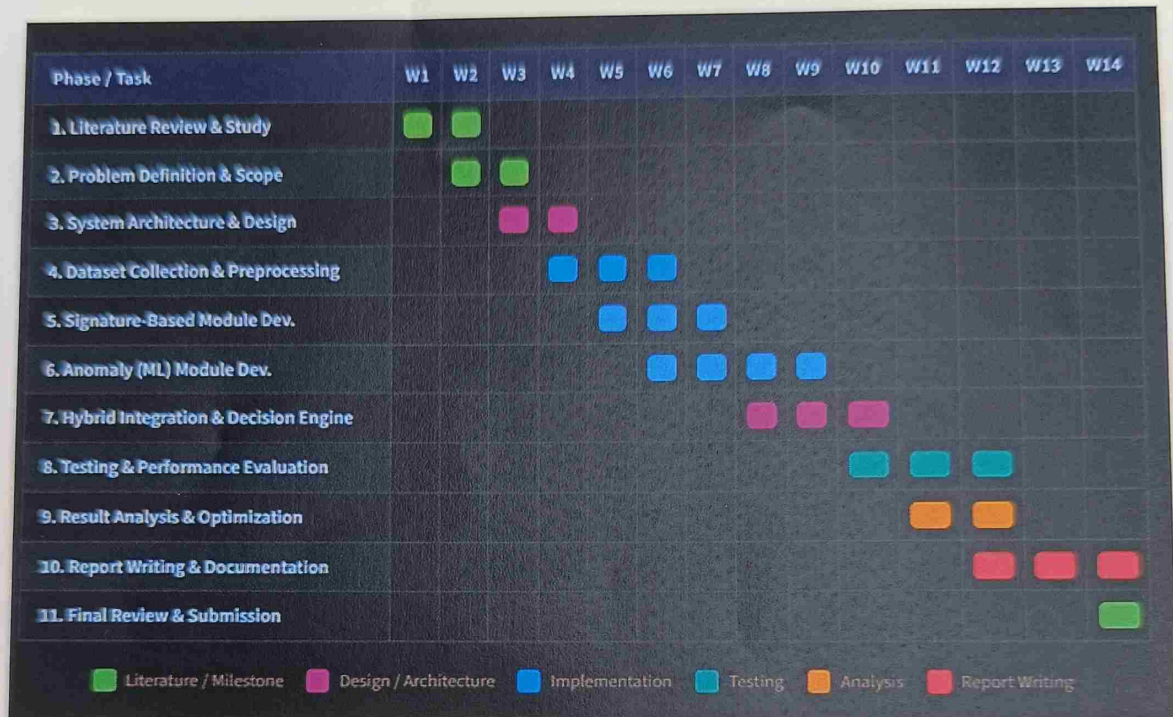


Figure 3. *Gantt chart*



Review of literature

Essential Literatures reviewed	Themes drawn from the Literatures
Studies on deep learning based IDS (Henry 2023, Sun 2020, Qazi 2023)	Combining CNN with RNN/GRU/LSTM improves detection accuracy up to 98–99% but struggles with rare attack types and dataset imbalance
Research on hybrid ML + DL approaches (Sajid 2024, Liu 2021, Duan 2025)	Multi-layer hybrid models achieve higher accuracy but at the cost of high computational load, making real-time deployment difficult
Optimized and ensemble based IDS (Verma 2025, Verma & Kumar 2026)	PSO and ensemble classifiers improve performance but lack real-world live network validation
IoT and domain specific IDS (Sharma 2025)	Hybrid encryption with ML works for IoT security but scalability remains a challenge in large networks
Systematic review of hybrid IDS (Maseno 2022)	No unified standard framework exists; false alarm reduction remains an open challenge across all reviewed hybrid approaches

Research Gaps:

- Existing hybrid models show poor detection for rare and minority attack types
- High computational cost makes most models unsuitable for real-time deployment
- Most studies are validated only on offline datasets, not on live networks
- Anomaly-based systems generate high false alarm rates, incorrectly flagging normal traffic as malicious, which reduces system reliability
- No standardized framework exists for integrating signature-based and anomaly-based detection in a single pipeline

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